

How the Proof Works

We are asking whether a viscous fluid that starts smooth can ever stop being smooth. This proof answers no. Start with a divergence-free velocity u_0 on $\mathbb{R}^3$ that is smooth and rapidly decaying with all its derivatives; this is the datum class $\mathcal{S}_\sigma$. Keep the viscosity $\nu > 0$ fixed. The difficulty is that viscosity is not the only part of the equation changing the fluid. The fluid also carries and deforms its own velocity:

$$\underbrace{\partial_t u}_{\text{local acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = - \underbrace{\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

The nonlinear term can sharpen the flow while viscosity smooths it. Pressure keeps the motion incompressible and couples every local change to the surrounding field. The regularity problem is decided by whether nonlinear sharpening can outrun viscous smoothing before some finite time T_* .

The place to see that danger is a vortex being stretched. Picture the surrounding flow pulling it lengthwise. Incompressibility conserves the volume of a material tube, so its cross-section contracts as its length increases. The core narrows. If the velocity change across that core does not fall just as quickly, then

$$|\nabla u| \sim \frac{|\Delta U|}{\ell}$$

grows as the radius ℓ shrinks. The velocity changes more sharply across less distance. This is nonlinear sharpening in the fluid itself. Viscosity resists it by diffusing velocity across the narrowing core.

Now the mathematics can name the contest. Write

$$\omega := \nabla \times u, \quad S := \frac{1}{2}(\nabla u + (\nabla u)^\top).$$

Taking the curl of Navier–Stokes gives

$$(\partial_t + u \cdot \nabla)\omega = S\omega + \nu \Delta \omega.$$

When the vorticity points along an extensional direction of S , the stretching term $S\omega$ strengthens it. This is how axial stretching can intensify rotation inside the narrowing core. Viscosity acts against that sharpening through $\nu \Delta \omega$. After integration over space, the contest is visible in one line:

$$\frac{1}{2} \frac{d}{dt} \|\omega(t)\|_2^2 = \int_{\mathbb{R}^3} \omega \cdot S\omega \, dx - \nu \|\nabla \omega(t)\|_2^2.$$

The first term is nonlinear production. The second is viscous removal.

Nothing here requires the kinetic energy itself to become infinite. In fact,

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \nu \|\nabla u(t)\|_2^2 = 0.$$

The total energy decreases. The unresolved possibility is concentration: finite energy can be carried by thinner regions while the velocity gradients inside those regions grow. A vortex can be losing energy and still be sharpening near its core.

So we shrink the picture and ask whether the balance changes. For $\lambda > 0$, set

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Every part of the equation picks up one common factor:

$$\begin{aligned}\partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), & \nu \Delta u_\lambda &= \lambda^3 \nu (\Delta u)(\lambda x, \lambda^2 t).\end{aligned}$$

Viscosity does not acquire an extra power of λ that makes it win automatically in a smaller vortex. Nonlinear transport, pressure, local acceleration, and diffusion remain matched by scaling.

The energy balance can miss precisely this concentration. Under the rescaling, velocity squared contributes λ^2 , while spatial volume contributes λ^{-3} , giving

$$\|u_\lambda(\cdot, t)\|_2^2 = \lambda^{-1} \|u(\cdot, \lambda^2 t)\|_2^2.$$

The concentrating copy becomes smaller in energy while its spatial changes become steeper. A spatial derivative contributes another factor λ , so measuring derivatives changes our sensitivity to the narrowing core. We want the height at which those effects cancel. The Sobolev norm $\|\cdot\|_{\dot{H}^s}$ measures s spatial derivatives in L^2 , including fractional orders. Its scaling is

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s},$$

concentration lowers the norm below $s = \frac{1}{2}$, raises it above $s = \frac{1}{2}$, and leaves it unchanged at

$$s = \frac{1}{2}.$$

This is the critical height. It is not a claim that nonlinear production and viscous dissipation are numerically equal at each moment. It says that scale alone gives neither one an advantage there.

Now keep following the vortex as its core narrows. If its radius is ℓ , a second spatial derivative across that core has size ℓ^{-2} relative to the velocity change it sees. The viscous term acts on the corresponding interval

$$|\nu \Delta u| \sim \frac{\nu}{\ell^2} |u|, \quad \tau_\ell \sim \frac{\ell^2}{\nu}.$$

If the core kept narrowing through $\ell_j = \rho^j \ell_0$ with $0 < \rho < 1$, those intervals would satisfy

$$\sum_{j=0}^{\infty} \frac{\ell_j^2}{\nu} = \frac{\ell_0^2}{\nu} \sum_{j=0}^{\infty} \rho^{2j} = \frac{\ell_0^2}{\nu(1-\rho^2)} < \infty.$$

That is the Zeno picture inside the vortex: each thinner core asks the equation to respond faster, and the resulting intervals can fit below one finite terminal time. It does not assert that a solution follows this cascade. It shows the failure pattern the proof has to prevent.

To turn this possible compression into a contradiction, suppose that the smooth solution has a finite maximal lifespan T_* . The scale calculation also points to another critical way to measure velocity in three dimensions:

$$\|u_\lambda(\cdot, t)\|_{L^3} = \|u(\cdot, \lambda^2 t)\|_{L^3}.$$

The L^3 norm, like $\dot{H}^{1/2}$, does not change under concentration. The endpoint continuation theorem now gives the decisive fork. If

$$\sup_{0 < t < T_*} \|u(t)\|_{L^3} < \infty,$$

then the solution continues smoothly through T_* . That is impossible if T_* is maximal, so the contrapositive requires

$$\sup_{0 < t < T_*} \|u(t)\|_{L^3} = \infty.$$

Here velocity blow-up means escape of the global critical L^3 norm; it does not mean that every particle speed must diverge.

The first finite-time integral keeps the physical meaning visible:

$$u(t) = u_0 + \int_0^t \partial_s u(s) ds.$$

If the local acceleration $\partial_t u$ stayed uniformly bounded in L^3 , this integral would keep the velocity bounded on the finite interval. So velocity escape forces the local acceleration to escape. The next integral is

$$\partial_t u(t) = \partial_t u(0) + \int_0^t \partial_s^2 u(s) ds.$$

If the jerk $\partial_t^2 u$ stayed bounded, acceleration would stay bounded and the first integral would again keep velocity bounded. This is the velocity–acceleration–jerk insight in plain language. In order for a singularity to form, velocity has to blow up. For that to happen, its acceleration also has to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

These responses are not added from outside. Acceleration is already written into Navier–Stokes. Substitute the equation for it and differentiate: jerk appears. Differentiate again and the next response appears. The first three rungs are

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u \\ &\quad - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The complete pattern is

$$\partial_t^{r+1} u = \nu \Delta \partial_t^r u - \sum_{a=0}^r \binom{r}{a} (\partial_t^a u \cdot \nabla) \partial_t^{r-a} u - \nabla \partial_t^r p.$$

The two finite-time integrals now repeat at every finite order j :

$$\partial_t^j u(t) - \partial_t^j u(0) = \int_0^t \partial_s^{j+1} u(s) ds,$$

and hence

$$\|\partial_t^j u(t)\|_3 \leq \|\partial_t^j u(0)\|_3 + \int_0^t \|\partial_s^{j+1} u(s)\|_3 ds.$$

The smooth datum makes every finite initial response finite. If one higher response stayed uniformly bounded on the finite interval, the response below it would stay bounded as well. Beginning with the required escape of u and repeating this implication upward gives, for every fixed j ,

$$\sup_{0 < t < T_*} \|\partial_t^j u(t)\|_3 = \infty.$$

Only now do we compress the notation:

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

Pressure has to travel with every response. Taking the divergence of Navier–Stokes and using $\nabla \cdot u = 0$ gives

$$-\Delta p = \partial_i \partial_j (u_i u_j).$$

Define the Riesz transform by

$$R_i := \partial_i (-\Delta)^{-1/2}.$$

With the decaying pressure normalization and repeated spatial indices summed, this convention gives $p = R_i R_j (u_i u_j)$. Differentiating that product r times recovers the pressure response from the velocity responses. The compact tower is

$$\begin{aligned} Q_r &= R_i R_j \sum_{a=0}^r \binom{r}{a} V_{a,i} V_{r-a,j}, \\ V_{r+1} &= \nu \Delta V_r - \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} - \nabla Q_r. \end{aligned}$$

Now the connection to spatial smoothness is visible. A hypothetical singularity runs forward through rougher and faster temporal responses. The proof reads the viscous relation backward. Once a finite response and its transport–pressure terms are controlled, the equation isolates ΔV_r and returns two spatial derivatives to the preceding response. Repeating a finite number of those controlled steps would raise the Sobolev height available to the original velocity. Higher temporal order points toward higher spatial order, but making that reversal useful requires bounds for the full signed interaction carried at each step.

That brings us back to the contest that opened the section. At critical height, the energy pairing separates nonlinear production from viscous removal without deleting either one. Let

$$\Lambda := (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

Pairing Navier–Stokes at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1},$$

where the complete signed production is

$$\mathcal{P}_s := -\langle \Lambda^s u, \Lambda^s ((u \cdot \nabla) u + \nabla p) \rangle.$$

Repeated differentiation at $s = \frac{1}{2}$ shows how viscous removal exposes the next spatial energy while the signed production differentiates alongside it:

$$\begin{aligned} H' &= \mathcal{P}_{1/2} - 2\nu E_{3/2}, \\ H'' &= \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}, \\ H''' &= \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}. \end{aligned}$$

At every finite order,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The n -th temporal derivative has exposed the spatial energy at height $n + \frac{1}{2}$, together with the production terms that must be controlled. The calculation lifts directly to every temporal response:

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2, \quad \partial_t E_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

Now the reversal has its payoff. Together with the L^2 energy already controlled, a bound at a higher spatial energy gives the corresponding H^s bound. For any requested classical derivative order k , it would be enough to reach

$$s > k + \frac{3}{2}, \quad H^s(\mathbb{R}^3) \hookrightarrow C_b^k(\mathbb{R}^3).$$

If every finite spatial height can be controlled, then no finite derivative order is left out:

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The temporal cascade that seemed to demand worsening behavior has exposed a route to complete spatial smoothness. This is the insight; its condition is still to be proved. The differentiated identities have exposed coordinates, not made them finite by rearrangement. Fix a finite temporal depth N and spatial height M . One step in the recurrence advances the temporal response by one and passes through two spatial derivatives in ΔV_r . That is why the finite estimate pairs height $M + 1$ of V_r with height $M - 1$ of V_{r+1} .

The whole-terminal rectangle theorem proves the required bound:

$$\mathfrak{R}_{N,M}(T) := \sum_{r=0}^N \|V_r\|_{L^2(0,T;H^{M+1})}^2 + \sum_{r=0}^N \|V_{r+1}\|_{L^2(0,T;H^{M-1})}^2,$$

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(T) < \infty.$$

Here N and M are fixed before T approaches T_* . This is the bound that survives every cutoff; smoothness on each shorter interval would not supply it. Let $T \uparrow T_*$. Each term is an integral of a nonnegative squared norm, so monotone convergence gives, for $0 \leq r \leq N$,

$$V_r \in L^2(0, T_*; H^{M+1}), \quad V_{r+1} \in L^2(0, T_*; H^{M-1}),$$

and the pressure formula gives the finite companion estimate

$$\|Q_r\|_{L^1(0,T_*;H^M)} \leq C_{M,r} \sum_{a=0}^r \|V_a\|_{L^2(0,T_*;H^{M+1})} \|V_{r-a}\|_{L^2(0,T_*;H^{M+1})} < \infty.$$

This uses the Riesz-transform bound, finite-order Sobolev products, and Cauchy–Schwarz in time. Pressure remains inside the finite record. The adjacent velocity pair then gives the middle trace:

$$V_r \in C([0, T_*]; H^M).$$

This is one finite rectangle. Its constant may depend on N and M . The proof never asks one constant to control infinitely many derivatives at once. It proves the required statement for every finite rectangle.

Those rectangles are not unrelated solutions. Every coordinate belongs to one velocity history generated by u_0 . When two rectangles overlap, their shared coordinates agree. Compatibility lets us assemble the entire family:

$$u_0 \longrightarrow \{\mathcal{R}_{N,M}[u_0; T_*]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] := \bigcap_{r,m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)), \quad u \in \mathcal{I}[u_0].$$

No infinite-order estimate has been smuggled into this step. Each coordinate came from a finite theorem; the intersection records that all of them belong to one compatible fluid.

Now take the zeroth temporal row. For every finite m , the rectangle pair and the equation place

$$u \in C([0, T_*]; H_\sigma^m).$$

All of these traces meet at one terminal value,

$$u_* := u(T_*) \in \bigcap_{m < \infty} H_\sigma^m = H_\sigma^\infty \hookrightarrow C^\infty.$$

At T_* , the equation converts the completed spatial information back into time information. Starting from $V_0^* = u_*$, the pressure formula gives Q_0^* , and the recurrence gives V_1^* . Repeating this at every finite order produces the pressure-complete terminal jet. Smooth local existence restarts from u_* ; uniqueness joins that continuation to the original history. This extends the solution past T_* , contradicting finite maximality. Hence

$$T_* = \infty.$$

After continuation, fix any finite time T and write $I = (0, T)$. The construction on that interval gives

$$u \in \mathcal{I}_T[u_0] := \bigcap_{a, m < \infty} H^a(I; H^m(\mathbb{R}^3)).$$

The familiar estimates are read from this finite-interval intersection instead of proved one by one. Every time space below refers to I , not to the whole infinite future. Given a finite response order r , a spatial derivative order k , and $2 \leq p \leq \infty$, choose

$$m > k + \frac{3}{2} - \frac{3}{p}.$$

That finite coordinate preserves the requested response order:

$$\begin{aligned} u \in \mathcal{I}_T[u_0] &\implies u \in H_t^{r+1} H_x^m \implies V_r \in H_t^1 H_x^m \\ &\implies V_r \in C_t H_x^m \implies V_r \in L_t^q W_x^{k,p} \quad (1 \leq q \leq \infty). \end{aligned}$$

At $r = 0$, the endpoint embedding $H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$, followed by the strict embeddings above with $m = 2$ and $m = 3$, recovers, among many others,

$$u \in L_t^\infty L_x^6, \quad u \in L_t^\infty L_x^\infty, \quad u \in L_t^\infty W_x^{1,\infty}.$$

These are not separate achievements. Each is one finite readout of $\mathcal{I}_T[u_0]$.

The whole motion is

$$u_0 \longrightarrow \{(V_r, Q_r)\}_{r \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \longrightarrow u_* \in H^\infty \longrightarrow \text{restart} \longrightarrow T_* = \infty.$$